

HydroServer-WIS2Box Data Workflows and Data Readiness for the Pilot

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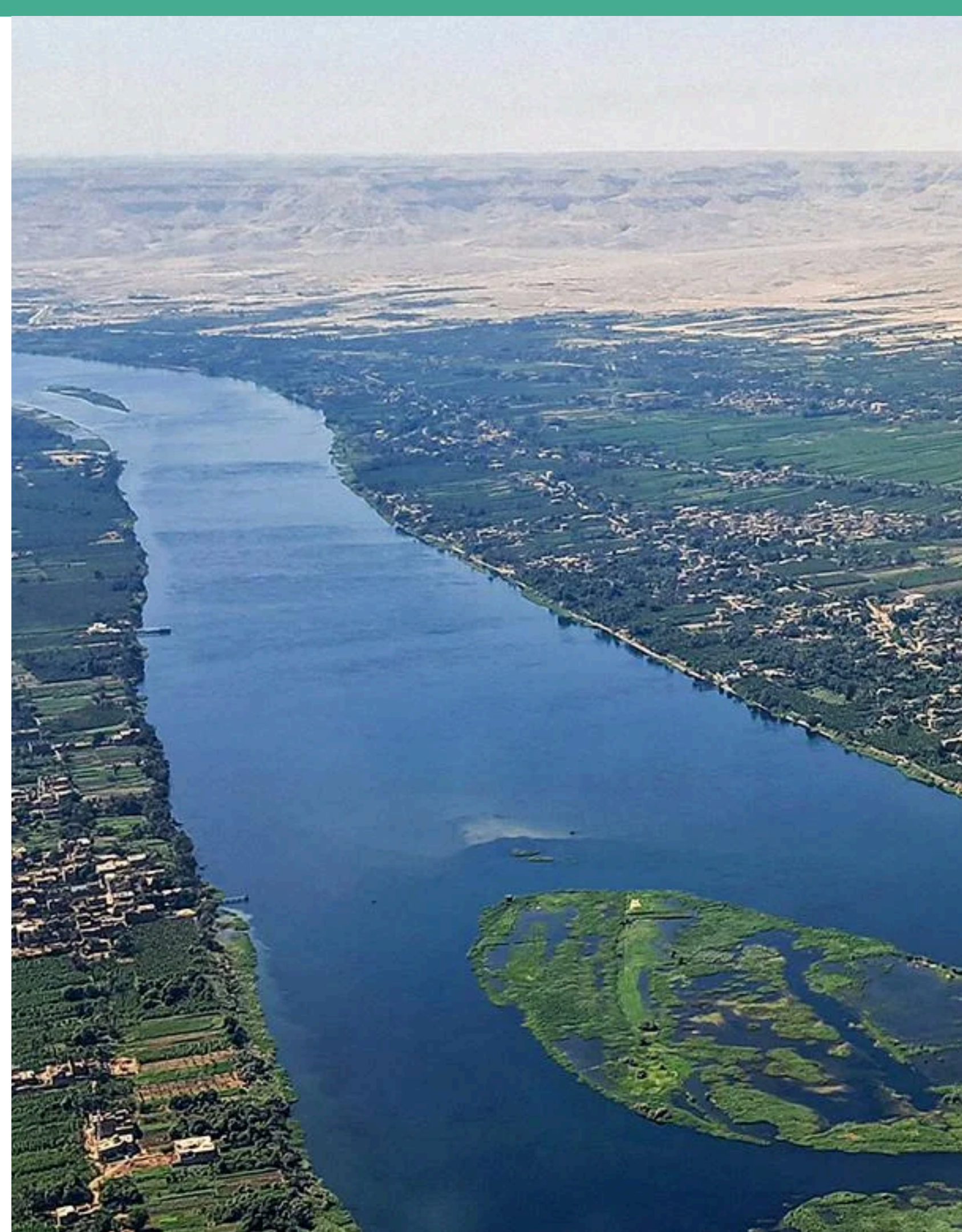
Session 8

Tuesday, September 16, 2026

09:00– 10:40



This workshop is supported with funding from the United States government.



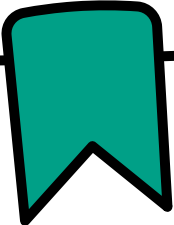
Yesterday, you received training on **WIS2Box** for **meteorological services** and **HydroServer** for hydrological services.



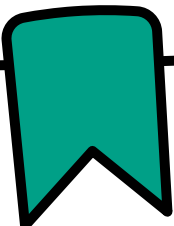
Today, we are going to explore some **potential workflows** showing how meteorological and hydrological services can use and exchange data between these systems

WIS2 & HydroServer Data Exchange

Just as recap...

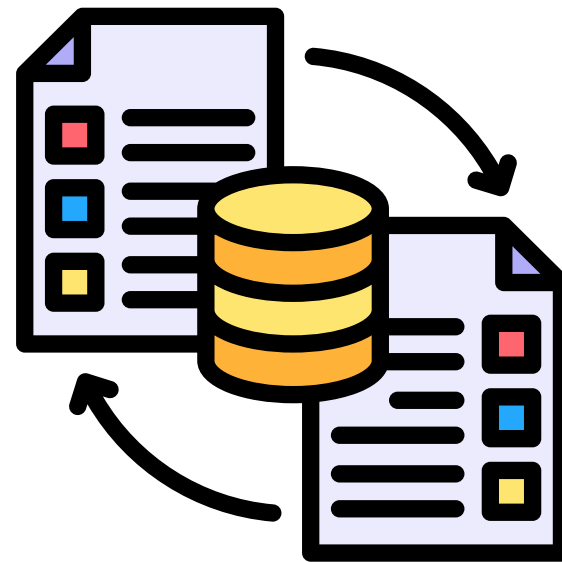


WIS2 is primarily an
exchange/distribution system.



HydroServer is more of a hydrologic **data**
management and publication system

Although **WIS2** and **HydroServer** have different primary purposes, they share important principles: both use **open standards**, are **free** and are designed to **facilitate data access, exchange, and use**.



Because their roles are complementary, there are some ways in which data can be exchanged between the two systems.

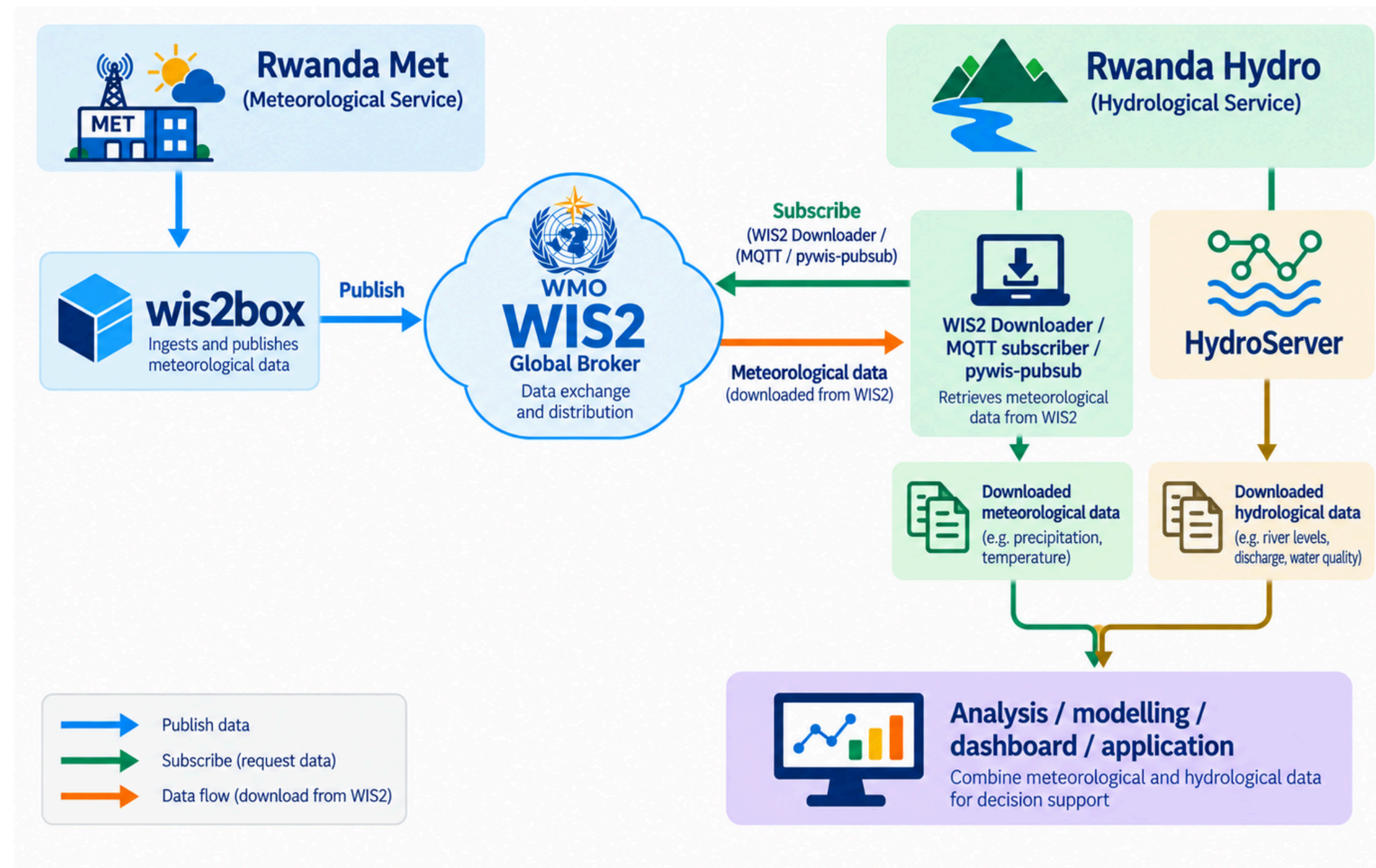
Option 1: Hydro Services Retrieve Meteorological Data from WIS2

In this example, **Rwanda Met** uses **WIS2Box** to **publish** meteorological data to **WIS2**.

Rwanda Hydro can then **subscribe** to **WIS2** and automatically retrieve the meteorological data it needs.

At the same time, **Rwanda Hydro** continues to manage its hydrological data in **HydroServer**.

The meteorological data from **WIS2** and the hydrological data from **HydroServer** can then be combined in an application, dashboard, or modelling workflow for analysis and decision support.

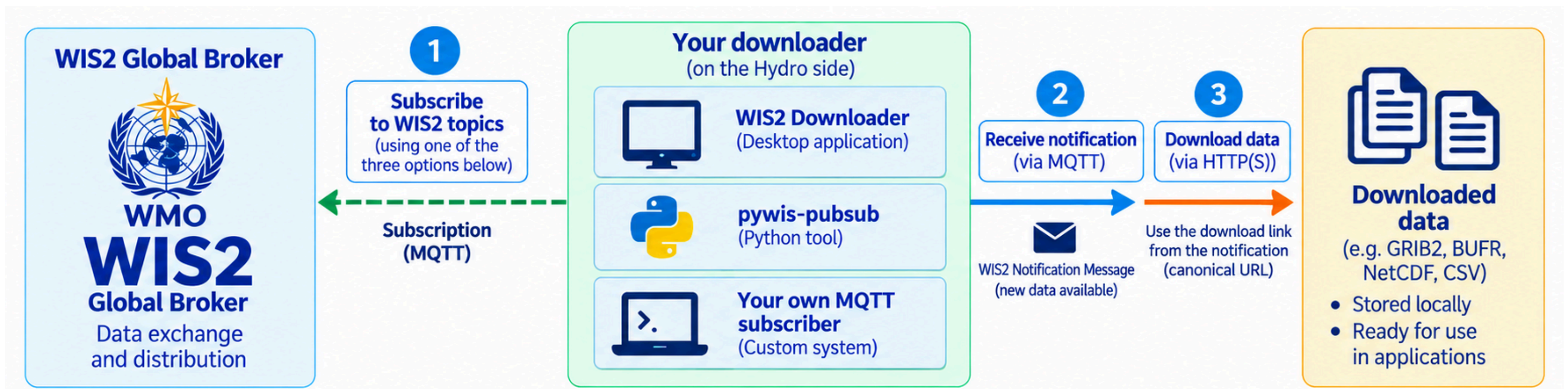


NOTE: WIS2 can include many different types of data, such as weather observations, hydrological data, satellite data, radar data, climate data, forecasts, and warnings. You can select and subscribe to the data you are interested in. Data published by Rwanda Met is just one example used in the diagram.

Ways to download data in WIS2

Rwanda Hydro can subscribe to **WIS2** using the WIS2 Downloader, a python tool, or a custom MQTT subscriber.

When matching data become available, a WIS2 notification is received and the data can be automatically retrieved.



*MQTT: Message Queuing Telemetry Transport

Option 2: Publishing Hydrological Data to WIS2

In this option, **Rwanda Hydro** publishes its hydrological data to **WIS2** using **wis2box**.

Depending on the data, the **Universal plugin** can publish files in their original format, while **data-specific plugins** can process, validate, or transform the data before publication, such as converting CSV observations to BUFR.

Once published, the data can be discovered and accessed by other WIS2 users, systems, and services.



This is currently available only for meteorological data (CSV to BUFR)

Option 2: Publishing Hydrological Data to WIS2

Universal plugin: publishes the original file without transforming its contents (e.g., CSV → CSV).

The data are **still discoverable (metadata)** and accessible through WIS2, but **users need to understand and handle the original format and structure themselves.**

People will have access to the metadata and a link to where the data is located.

Manual and Telemetry Data: Key Considerations for the Pilot

These are **key considerations** to **review** when **planning the ingestion of manual or telemetry station data for the pilot**, to ensure a successful ingestion workflow.



Remember: Keep your plan achievable! It's better to start with fewer stations than to include too many.

Manual Data – Readiness Checklist

- ① Identify priority datasets and stations - historical and current/near-real-time.
- ② Identify where the data are stored - spreadsheets, CSV files, databases, etc.
- ③ Define who is responsible for uploading the data.
- ④ Define upload frequency - daily, weekly, monthly, etc.
- ⑤ Standardize data and metadata - station IDs, variables, units, timestamps.
- ⑥ Perform a data quality and completeness assessment - missing values, gaps, errors, etc.
- ⑦ Establish an ingestion workflow for future uploads

Telemetry Data – Readiness Checklist

- 1 Identify priority datasets and stations.
- 2 Assess station connectivity – cellular, radio, satellite, Internet.
- 3 Identify data access method – API, telemetry platform, FTP/SFTP, etc.
- 4 Define transmission frequency – continuous, hourly, daily, etc.
- 5 Ensure local buffering/storage when connectivity is lost.
- 6 Enable retransmission when connectivity is restored.
- 7 Standardize incoming data and metadata.
- 8 Define automated quality-control requirements
- 9 Establish and monitor the ingestion workflow and system performance.

Any Question?

